

Complexity is a unified cognitive kind

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Abstract (250 words)

Any information-processing system must deal with the complexity of its input. In humans and other organisms, the pressure to meet this challenge is so great that it is often placed at the center of evolutionary accounts of cognition. However, complexity arises in many different forms: A visual image may be complex, an auditory sequence may be complex, and so too for linguistic or mathematical expressions. Are these disparate forms of complexity ‘unified’ in the mind? In other words, does cognition represent a common quantity shared across qualitatively different types of information? Longstanding proposals for the format of cognition, reinvigorated by recent experimental evidence, point to this possibility. Here, 9 experiments reveal domain-general representation of complexity. Experiments 1–7 used a reward-transfer task to reveal that outcomes associated with complexity generalize across domains. For example, having learned that more jagged shapes give higher rewards, subjects spontaneously inferred that more sophisticated melodies give higher rewards as well. Such transfer arose across a wide range of stimuli, including dot-arrays, mathematical expressions, letter-strings, and tactile forms. Experiment 8 demonstrated that such transfer occurs automatically, intruding on task-irrelevant judgments. Finally, Experiment 9 revealed reliable individual differences in higher-level judgments across stimulus domains; for example, subjects who find simple shapes more aesthetically pleasing than complex shapes also find simple melodies more pleasing than complex melodies. Together, these results suggest that the mind represents information density as a type-independent quantity, building upon recent hypotheses of an underlying representational language shared across cognitive domains.

Main text

All intelligent systems grapple with the complexity of the information they process. In humans, non-human animals, and information-processing machines more generally, meeting this challenge requires efficiently representing complex inputs, often in the service of perception, action, learning, and other downstream processes. This demand is so fundamental that it is often placed at the center of evolutionary accounts of cognition, with some theorists proposing that “the function of cognition is to enable the agent to deal with environmental complexity” (Godfrey-Smith, 1998; Godfrey-Smith, 2013).

Complexity poses a cognitive challenge not only because complex stimuli contain large amounts of information, but also because different stimuli may be complex for different reasons. A geometric shape, a sequence of tones, or a linguistic or mathematical expression can each be complex, but different features determine the complexity of these very different stimuli. For example, the complexity of a shape may be defined by its number of turns (Attneave, 1957), whereas the complexity of a sequence of tones may be defined by the variety of its rhythm (Bouwer et al., 2018). Are these disparate examples unified in any meaningful way? Or do they reflect entirely separate notions of complexity that just happen to share the same name?

One clue that seemingly distant instances of complexity in fact share a common currency comes from computer science and information theory, where the discovery of *universal compression algorithms* demonstrated that complex information of any type or format can be compressed by the same procedure. Such algorithms compress data without regard for the data’s type or source distribution, facilitating efficient storage and recovery (as in ZIP compression; see Lempel and Ziv, 1976; Ziv and Lempel, 1978; Welch, 1984). The significance of this demonstration arises not just from its efficiency, but also from its *universality* (MacKay, 2003), applying to text, visual images, auditory sequences, or any arbitrary data format. This suggests that all complex stimuli, no matter their format, share a stimulus-invariant property that is (at least in principle) extractable by information-processing systems.

These observations raise a question for psychology: Does the mind appreciate a universal notion of complexity, abstracted away from any particular domain?

Complexity in the mind: One thing or many?

On the one hand, psychological studies of complexity take a wide range of experimental and theoretical approaches, in ways that seem to implicate a multitude of domain-specific notions of complexity. Complexity underlies many important aspects of cognition (for review, see Chater, 1999; Chater & Vitányi, 2003; Feldman, 2016); but research on cognitive representations of complexity typically focuses on one kind of complexity in isolation (cf. Boger & Keil, 2025). For example, in the realm of visual complexity — i.e., the complexity of a visual shape or pattern — previous work reveals that complexity drives memory (Alvarez & Cavanagh, 2004), attention (Sun & Firestone, 2021), aesthetic preferences (Osborne & Farley, 1970; Sun & Firestone, 2022a), word-labeling (Lewis & Frank, 2016), and even verbal descriptions (Sun & Firestone,

2022b). However, the specific notions of complexity explored in such work do not — and often cannot — interact with each other. The complexity of an image has been defined by its fractal dimension (Spehar et al., 2003); the complexity of a geometric shape has been defined by the structural surprisal of its shape skeleton (Sun & Firestone, 2021); the complexity of a labeled object has been defined by the number of geons it contains (Lewis & Frank, 2016); and so on. Beyond vision, the complexity of a concept has been defined in terms of Boolean compressibility (the length of the shortest logically equivalent propositional formula; Feldman, 2000; Feldman, 2003); the complexity of a tool has been defined in terms of functional diversity (Ahl et al., 2017); and the complexity of a social network has been defined in terms of its graph structure (Butts, 2001). These vastly different approaches to complexity in psychology — both across and even within domains — may promote theorizing of complexity as many different domain-specific quantities that only happen to share the same name.

On the other hand, there is an intuitive sense in which we can appreciate complexity *across* very different contexts. Consider Figure 1, which depicts complex and simple stimuli from many domains. Despite the variety of the stimuli, it may nevertheless feel that there is a single underlying quantity that unifies them — i.e., something that separates the top row from the bottom row even across their different domains and features. This raises the intriguing possibility that complexity is mentally represented in a more domain-general manner.

Recent experimental evidence suggests that the mind relies on a compression-like algorithm to represent stimuli *within* particular domains. For example, it has recently been proposed that a ‘language of geometry’ — wherein geometric patterns are deconstructed into primitive symbols such as line segments and curves recursively composed with operators such as ‘turn’ and ‘move’ — forms the basis for mental representations of shapes (Amalric et al., 2017; Sablé-Meyer et al., 2022). Such representations rely on faithfully compressing a shape into its minimal form (a capacity that may even reflect a signature of uniquely human reasoning; Sablé-Meyer et al., 2021). Similar kinds of compressed representations may arise in other cognitive domains, such as language, audition, and more (for review, see Dehaene et al., 2022), and these compressed representations predict performance on a variety of cognitive tasks (see, e.g., Mathy & Feldman, 2012; Al Roumi et al., 2017). Such proposals also interact with longstanding discussions of the format of mental representations, including the possibility that a more general ‘language of thought’ pervades the mind and underwrites performance in otherwise-disparate cognitive domains (Fodor, 1975; Quilty-Dunn et al., 2023). This picture of mental organization would naturally enable exactly the sort of cross-domain interaction (e.g., of geometry and tones, or language and math) that we explore here.

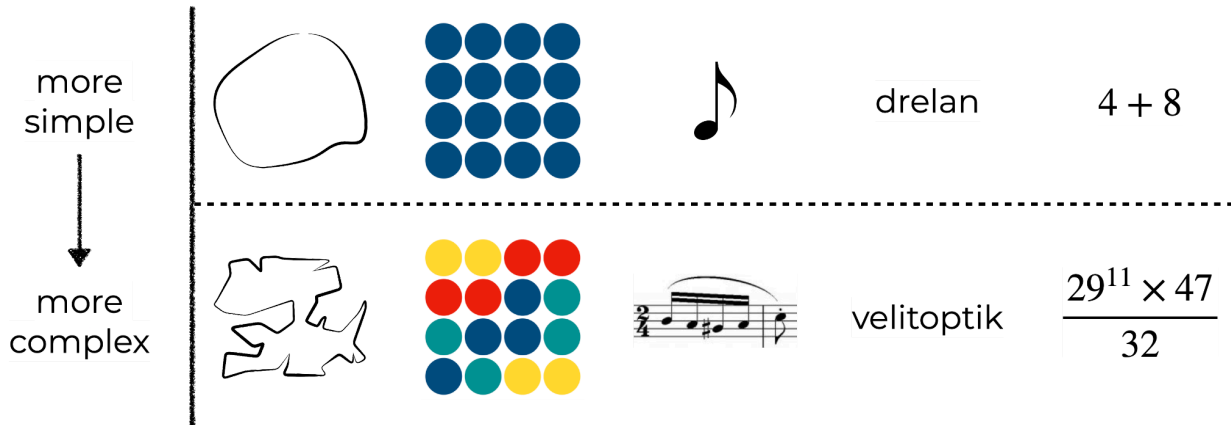


Fig 1. Simple and complex stimuli from a variety of domains. The features that give rise to complexity are qualitatively different across domains — e.g., the complexity of a geometric shape may be defined in terms of its number of turns, while the complexity of a mathematical expression may be defined in terms of the operations it contains. The present work asks whether the mind extracts and represents a shared property across these domains.

The present experiments: Complexity transfer across domains

Here, we ask whether the mind represents complexity in a domain-general manner. Our approach is to test whether representations of complexity *transfer* from one domain to another. In other words, we ask: If some pattern is associated with complexity in one domain, will that same pattern transfer to a completely different domain? This approach allows for the study of many different kinds of complexity at once, in contrast to previous studies that typically examine a single type of complexity in isolation. We conducted 9 studies that explore this question using three different paradigms. Experiments 1–7 demonstrated that information a subject acquires about complex stimuli in one domain informs their judgments about complex stimuli in other domains, spanning visual, auditory, linguistic, mathematical, and tactile modalities. Experiment 8 showed that such transfer occurs automatically, such that incidental exposure to task-irrelevant complex stimuli in one domain intrudes on basic judgments made in another domain. Finally, Experiment 9 revealed stable individual differences in aesthetic judgments of complexity across domains, in ways that seem explicable only by a domain-independent, individual-level preference for information density of a certain degree. Together, these results suggest that the mind possesses a unified representation of complexity abstracted away from the surface features of any given stimulus, opening the door to questions concerning the evolutionary, developmental, and computational foundations of complexity representation.

Experiments 1–7: Reward-transfer across domains

Our first set of experiments (Experiments 1–7) leveraged a reward-transfer task to ask whether learned associations with complexity transfer spontaneously across domains. We designed a reward-learning task consisting of two parts. In the first part of the task (the learning phase), participants completed 12 trials in which they chose between a simple and complex stimulus from a given domain (e.g., a simple geometric shape vs. a complex geometric shape).

Participants were consistently rewarded for choosing either the simple or complex option (randomized between-subjects), allowing them to associate rewards with either simple or complex stimuli in a given class. Then, in the second part of the task (the transfer phase), participants chose between a simple and complex stimulus of a novel, unseen domain. Participants had never seen this new stimulus class (and thus had never learned a reward-rule to associate with it); they were never instructed about the nature or generality of the reward-rule; and they received only one trial of this novel stimulus class. Would learned associations with complexity generalize to a totally new stimulus class? In other words, having learned to associate greater rewards with choosing complex (or simple) stimuli of one kind (e.g., geometric shapes), would participants spontaneously choose complex (or simple) stimuli of another kind (e.g., dot-arrays, melodies, letter-strings, mathematical formulas, and so on)?

Experiment 1 tested bidirectional transfer between geometric shapes and dot-arrays (Figure 2A). Though both stimuli are visual, they are complex for seemingly different reasons: A shape's complexity may be determined by (say) its number of turns, whereas a dot-array's complexity may be determined by the number (or arrangement) of its colors. By trial 3 of the learning phase, where subjects received feedback on their selections, performance was 90%; by trial 12 performance was 99%. This suggests that subjects successfully learned an implied association between complexity and reward. Our key question, however, concerns the *generality* of this learned rule; how far does it extend? On the crucial transfer trial, when subjects were shown a stimulus pair from the class they had not been exposed to (e.g., dot-arrays after seeing geometric shapes), 86.3% of subjects selected the reward-earning stimulus ($p < .001$ in exact binomial test relative to chance); e.g., they readily selected the more variable dot-array after having learned that more jagged shapes were rewarded. Thus, subjects did not merely learn a local reward-rule for a given stimulus class (e.g., to choose the shape with more turns, or the dot-array with more colors). Rather, they learned a general reward-rule that transferred across stimulus classes (i.e., to choose the more complex stimulus), even in this one-shot context.

Still, shapes and dot-arrays are both visual images, so it may not be surprising to observe transfer across the two. Subsequent experiments thus generalized this pattern much further, testing a very wide range of stimuli. Experiment 2 found successful one-shot transfer between geometric shapes and melodies (at a rate of 71.9%, $p < .001$; Figure 2B); Experiment 3 found successful one-shot transfer between geometric shapes and letter-strings (at a rate of 78.0%, $p < .001$; Figure 2C); and Experiment 4 found successful one-shot transfer between geometric shapes and mathematical expressions (at a rate of 64.3%, $p < .01$; Figure 2D). In other words, having learned simply that more jagged shapes are worth more points, subjects selected longer melodies, words with more syllables, and more elaborate mathematical expressions. This is especially striking given that, across this broad range of stimuli, individual stimulus pairs not only arise from different modalities but also span from low-level notions of visual complexity to more high-level notions of cognitive or operational complexity. The features of each of these stimuli vary dramatically in type; and in almost every case, there is no obvious way to map these features across their disparate types. In other words, there is no reason to expect the mind to transfer notions of complexity from one domain (e.g., curvature of a geometric shape) to notions of complexity from other, very different domains (e.g., number of operators in a mathematical

expression). Thus, Experiments 1–4 point towards domain-general, one-shot transfer of complexity in a reward-learning task.

Experiment 5 provided further evidence for this transfer by testing all possible stimulus pairs from Experiments 1–4, simultaneously. Rather than testing bidirectional transfer between geometric shapes and other stimuli (dot-arrays, melodies, letter-strings, and mathematical expressions), Experiment 5 tested bidirectional transfer between *all* of these stimuli at once. Subjects were randomly assigned to receive any of the 5 stimulus classes (used in Experiments 1–4) in the learning phase. Then, the transfer phase consisted of one trial of each of the other 4 held-out stimulus classes (all without feedback). This allowed us to test the limits of this domain-general, asking whether transfer of complexity would arise from any stimulus class to any other stimulus class. Reflecting the generality of these representations, this experiment revealed near-universal transfer. First, successful transfer arose *from* every stimulus class; participants who received any of the 5 stimulus classes in the learning phase earned rewards at an above-chance rate in the transfer phase (mean transfer accuracy for participants who learned shapes: 74.3%, $t(111) = 9.56$, $p < .001$, $d = 0.90$; dot-arrays: 68.6%, $t(81) = 5.84$, $p < .001$, $d = 0.65$; melodies: 81.0%, $t(80) = 11.35$, $p < .001$, $d = 1.26$; letter-strings: 65.8%, $t(90) = 4.83$, $p < .001$, $d = 0.51$; mathematical expressions: 61.0%, $t(92) = 3.62$, $p < .001$, $d = 0.38$). Next, successful transfer arose *to* every stimulus class; participants who received a single trial of any of the 4 held-out stimulus classes in the transfer phase earned rewards at an above-chance rate (mean transfer accuracy for shape transfer trials: 78.0%, $p < .001$ in exact binomial test; dot-arrays: 73.7%, $p < .001$; melodies: 63.8%, $p < .001$; letter-strings: 65.5%, $p < .001$; mathematical expressions: 70.4%, $p < .001$). Finally, successful transfer arose *between* 9/10 possible bidirectional learning-transfer pairs, and approached significance in the 10th (dot-arrays and mathematical expressions: 64.0%, $p < .001$ in exact binomial test; dot-arrays and melodies: 75.5%, $p < .001$; dot-arrays and shapes: 82.5%, $p < .001$; dot-arrays and letter-strings: 62.2%, $p < .01$; mathematical expressions and melodies: 65.3%, $p < .001$; mathematical expressions and shapes: 75.6%, $p < .001$; mathematical expressions and letter-strings: 56.5%, $p = .09$; melodies and shapes: 74.1%, $p < .001$; melodies and letter-strings: 72.1%, $p < .001$; shapes and letter-strings: 71.8%, $p < .001$; Figure 2E). Thus, successful transfer arose from, to, and between stimulus classes, suggesting the mind capitalizes on a domain-general representation of complexity.

Experiment 6 controlled for a possible confound in these results concerning stimulus magnitude. When two stimuli differ in complexity, they must also differ in at least one stimulus magnitude dimension (e.g., curves in a shape, notes in a melody, etc.). Thus, perhaps our previous results point to the transfer of merely *any* stimulus magnitude, and not specifically those implicated in complexity. Note that it is difficult to articulate this reductionist explanation without invoking complexity itself; after all, a shape with ‘fewer turns’ has ‘more smoothness’, a word with ‘fewer syllables’ may have ‘more readability’, and so on. It is only because having more turns is *already* determinative of complexity that it is subject to transfer. Still, we designed an experiment that dissociated complexity from merely having ‘more’ of any arbitrary magnitude. Experiment 6 used Experiment 1’s design — testing bidirectional transfer between geometric shapes and dot-arrays — but included one key additional transfer trial where participants were presented

with two colors of identical hue and lightness, but of differing saturation (one 25% and one 75%). If the mind merely transfers a notion of ‘more’ from one stimulus class to another, then participants who are rewarded for choosing the more complex geometric shapes (or dot-arrays) should also choose the more saturated color. However, if the mind is transferring a deeper notion of complexity from one stimulus class to another, then participants should choose between the colors at random (while still succeeding on the other transfer trial, as in Experiment 1). The latter is exactly what we observed: Participants successfully transferred the reward-rule between geometric shapes and dot-arrays (84.4%, $p < .001$ in exact binomial test, replicating the results of Experiment 1), but were indistinguishable from chance in the color transfer trial (48.4%, $p = 0.84$; Figure 2F). (The difference in accuracy between the two transfer trials — i.e., the color transfer trial and the shape or dot-array transfer trial — was significant; $t(94) = 5.99$, $p < .001$, $d = 0.62$.) This suggests that the nature of the transfer relies on complexity itself, and not merely on a confounding stimulus magnitude dimension.

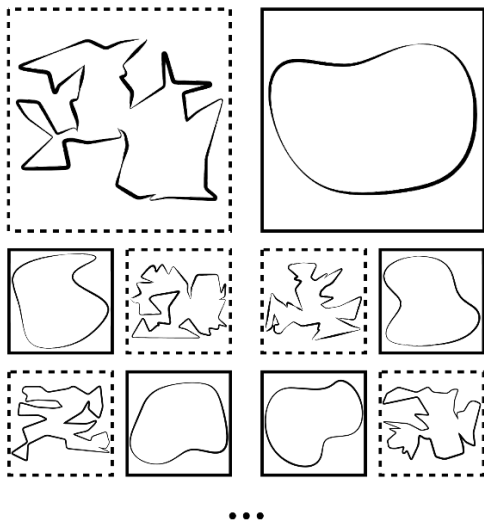
Finally, Experiment 7 generalized this pattern to an additional modality: touch. We 3D printed the geometric shapes used in previous experiments (using a Bambu Lab X1C printer) and asked whether transfer would arise between these tactile forms and the visual dot-arrays used earlier. As before, participants completed 12 learning trials of a randomly assigned stimulus class (either tactile forms or visual dot-arrays), and then a single transfer trial of the other class. Importantly, participants had no visual access to the tactile forms; the objects were placed inside a closed box with one hole on each side to allow participants to touch the shapes without looking at them. Still, we found successful transfer across these domains (at a rate of 81.8%, $p < .001$; Figure 2G). In other words, having learned to choose the complex tactile forms, subjects spontaneously selected the complex dot-arrays; thus, the observed transfer even extended across vision and touch.

Together, Experiments 1–7 demonstrate spontaneous reward-transfer of complexity across the visual, auditory, linguistic, mathematical, and tactile domains.

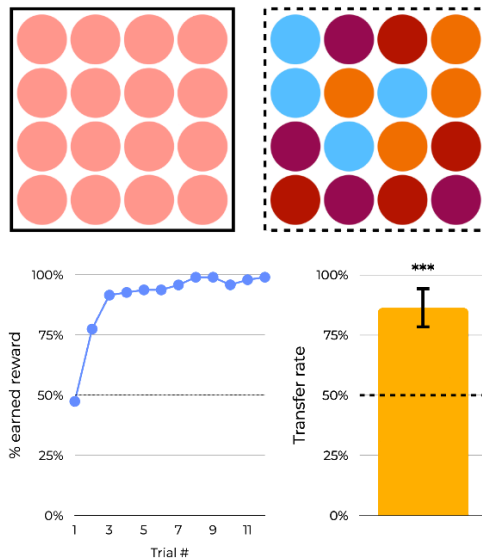
(A)

Learn**Transfer**

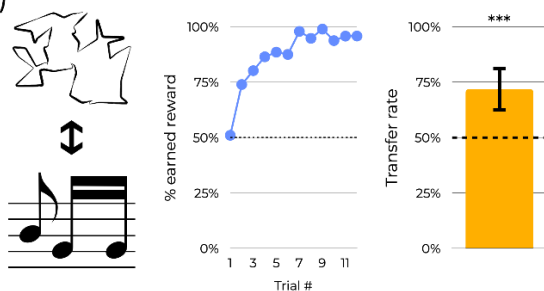
Task: Which option will earn you coins?
(12 trials with feedback)



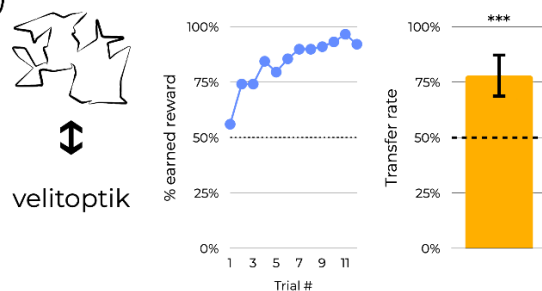
Task: Which option will earn you coins?
(1 trial without feedback)



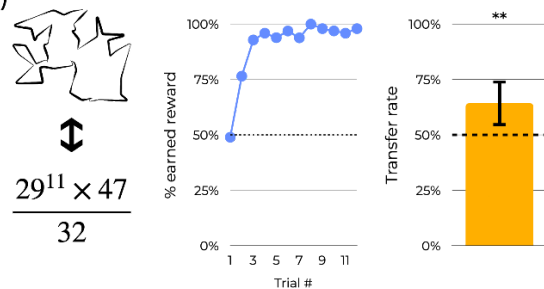
(B)



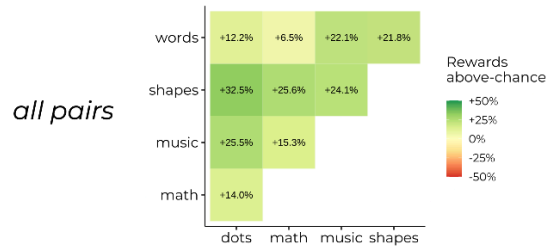
(C)



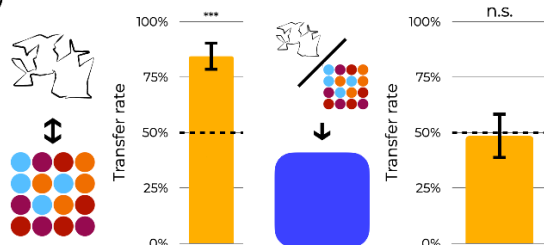
(D)



(E)



(F)



(G)

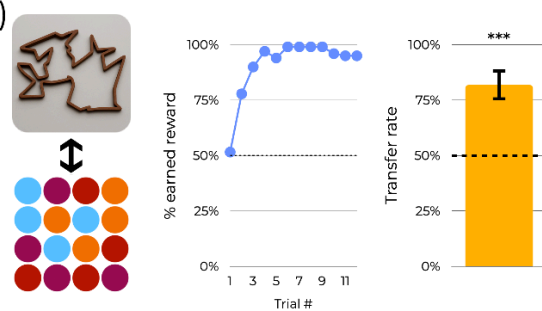


Fig 2. Experimental design and results of the reward-transfer tasks (Experiments 1–7). (A) In each of Experiments 1–7, participants completed 12 learning trials. Each trial contained one simple stimulus and one complex stimulus from a single domain. Participants chose between the two options and received feedback (i.e., reward or no reward), allowing them to learn the reward-rule (e.g., choosing the complex option earns you a reward). Then, participants saw one trial of a new stimulus class which they had never seen before. Would the learned reward rule transfer to this totally new stimulus class? Experiment 1 tested bidirectional transfer between geometric shapes and dot-arrays. Participants successfully learned the reward-rule, and then spontaneously transferred it to a previously unseen stimulus class. This suggests that the reward-rule operated over a general notion of complexity that applied to both kinds of stimuli, rather than a local reward-rule that applied only to one kind. (B) Experiment 2 tested transfer between geometric shapes and melodies, again revealing successful transfer between the two. (C) Experiment 3 found transfer between geometric shapes and letter-strings. (D) Experiment 4 found transfer between geometric shapes and mathematical expressions. (E) Experiment 5 tested all 5 stimulus classes used in Experiments 1–4 simultaneously. Subjects received a random stimulus class in the learning phase; then, in the transfer phase, they received a single trial (without feedback) of each of the other 4 held-out stimulus classes. We found successful transfer between all pairs of stimuli. (F) Experiment 6 ensured that this transfer does not arise for merely any stimulus dimension. This experiment replicated Experiment 1 but included an additional transfer trial where two colored patches were presented at different levels of saturation (25% and 75%). We replicated the transfer between the geometric shapes and dot-arrays (found in Experiment 1). But, neither stimulus type transferred to the colors, suggesting that the observed transfer is really about more (or less) *complexity* and not just any more (or less) of any arbitrary stimulus magnitude. (G) Experiment 7 tested transfer between 3D printed tactile forms and dot-arrays. Participants had no visual access to the tactile forms; yet, transfer arose across vision and touch. Readers can experience all the experiments presented in this paper (with the exception of Experiment 7, which was conducted in-person with tactile forms) at https://perceptionresearch.org/complexity_transfer.

Experiment 8: Automatic interference across domains

Experiments 1–7 revealed that the mind can — and does — transfer complexity across domains. However, the task used in those experiments may seem to actively invite such transfer: It essentially asks subjects to make an educated guess about which regularities might apply to the new stimuli they encounter. Could such transfer also be observed in a task where such mapping not only is unnecessary, but also may actively *impair* performance?

Experiment 8 tested whether complexity transfer arises *automatically*, asking whether task-irrelevant complex stimuli in one domain intrude upon judgments about complexity in another domain. Participants completed a modified Stroop task: On each trial, participants saw two words — one simple word and one complex word — and simply had to say which was more complex. However, above each word appeared a geometric shape that was either congruent to the word's complexity (i.e., complex shape above complex word, simple shape above simple word; half of trials) or incongruent to the word's complexity (i.e., complex shape above simple word, simple shape above complex word; half of trials; Figure 3A). Since participants simply had to judge which word was more complex, they could ignore the shapes entirely (and were explicitly told this). In other words, the shape was (a) completely irrelevant to the word task and (b) demanded no response of any kind.

Still, shape complexity automatically intruded on judgments of word complexity: Participants gave faster responses on congruent trials (57.4ms difference, $t(47) = 9.07$, $p < .001$, $d = 1.31$ in paired t -test), and this did not come at the expense of accuracy (indeed, accuracy was higher on congruent trials as well; +2.49%, $t(47) = 3.26$, $p < .01$, $d = 0.47$; Figure 3B). Put the other way around, *incongruence* between shape complexity and word complexity made subjects slower and less accurate, *impairing* their performance. Thus, even though participants could have safely ignored shape complexity and its relation to the stimuli in front of them (and indeed may have performed better if they had done so), they were simply unable to do so; complexity in one domain intruded on judgments of complexity in the other. This suggests that domain-general representations of complexity not only operate on explicit tasks (such as our reward-learning paradigm) but also transfer automatically in ways that interfere with other processes and judgments.

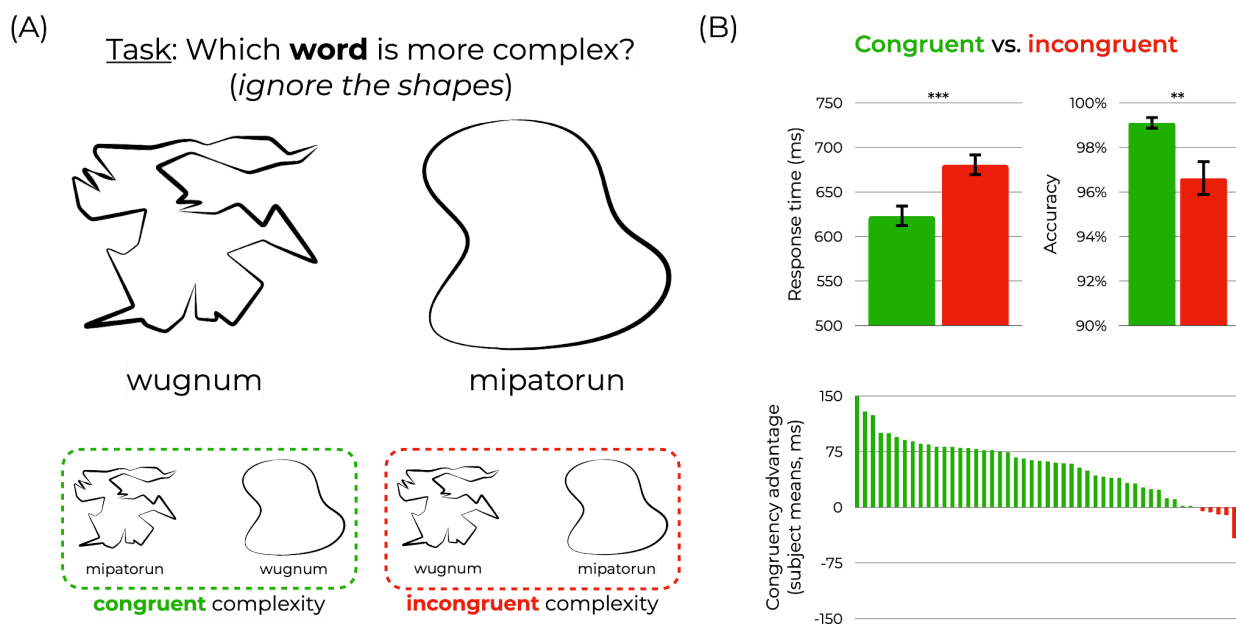


Fig 3. Complexity automatically transfers across domains. (A) Design of the novel Stroop-like task used in Experiment 8. Subjects were presented with two words and had to judge which was more (or less) complex. Above each word, a task-irrelevant shape was presented. Subjects were instructed to ignore the shapes, and indeed never had to make any judgments about them — thus, the shapes were entirely task-irrelevant. Importantly, in half of trials, shape complexity was congruent with word complexity (i.e., complex shape above complex word, simple shape above simple word); in the other half of trials, shape complexity was incongruent with word complexity (i.e., complex shape above simple word, simple shape above complex word). (B) Participants were faster and more accurate on congruent trials than incongruent trials, and almost every participant showed this congruency advantage. This result suggests that task-irrelevant representations of complexity in one domain may automatically interfere with basic judgments of complexity in another domain.

Experiment 9: Complexity preferences are stable across domains

Our previous experiments demonstrate that the mind transfers notions of complexity across domains, in ways that even intrude on basic and incidental judgments. But if complexity is represented as a domain-general quantity, might we find evidence that a single representation governs and unites downstream judgments that might otherwise be unrelated? Experiment 9 explored this question at the subject-level, asking whether an individual's aesthetic preferences for complex stimuli are correlated across domains.

Our approach in this experiment neither explicitly (as in Experiments 1–7) nor implicitly (as in Experiment 8) invited subjects to map stimuli of one domain to stimuli of another. Rather, this experiment explored whether judgments within each domain emanate from a more general type-independent preference that is stable within an individual. A large body of work finds strong relationships between complexity and aesthetics across a broad range of stimuli, including abstract art (Osborne & Farley, 1970), geometric shapes (Sun & Firestone, 2022a), melodies (Delplanque et al., 2019; Beauvois, 2007), and even food (Lévy et al., 2006). However, are complexity-driven aesthetic preferences consistent *across* domains? For example, if someone prefers to view especially complex geometric shapes, does that same person also prefer to hear especially complex melodies (Figure 4A)? If so, this would be consistent with a unified representation of complexity that gives rise to an individual 'information density' preference in any given domain.

Recent evidence points to the possibility that other sorts of aesthetic preferences are stable across domains (Chen et al., 2022). Here, we apply a similar approach to our questions about complexity. In Experiment 9, participants completed an aesthetic preference task. On each trial, participants were presented with two stimuli from a given domain (geometric shapes, dot-arrays, melodies, letter-strings, and mathematical expressions; as used in Experiments 1–5) and chose which one they found more (or less) beautiful. Importantly, across trials, the domains varied randomly; e.g., on one trial, a participant would choose between two geometric shapes, and on another, they would choose between two mathematical expressions (Figure 4B). This approach yielded individual subject-level aesthetic preferences in many different domains (Figure 4C).

The results of this task were consistent with a stimulus-independent preference that applied across domains. An intraclass correlation revealed consistent individual differences in complexity-driven aesthetic preferences across domains (two-way ICC(C,5) = 0.37, $F(149, 596) = 1.58$, $p < .001$; Figure 4D). In other words, the stronger a subject's preference for, e.g., complex shapes, the stronger their preference for complex dot-arrays, melodies, words, and mathematical expressions. Additionally, we computed two bootstrapped measures of consistency, asking whether subjects' preferences across domains were more consistent than what would be expected by chance. In each iteration of this test, we shuffled the values (or ranks) across subjects within a domain; then, we computed the new mean absolute error for each subject on these shuffled values. Finally, we compared the permuted errors to the real errors, and found that the observed cross-domain consistency was higher than the permuted cross-domain consistency in 1000/1000 iterations when considering the raw values (and

999/1000 iterations when considering the ranks; Figure 4E). Thus, individual complexity preferences were substantially more consistent across domains than what would be expected by chance.

Note that this result cannot be explained by a mere experimental demand to map complexity across domains. Even more so than in Experiment 8, such a demand was completely absent, especially given that (a) within a trial only a single domain appeared and (b) there was no reward to earn or even a correct answer to give. In other words, this experiment suggests that complexity drives stable individual differences in higher-level judgments across domains.

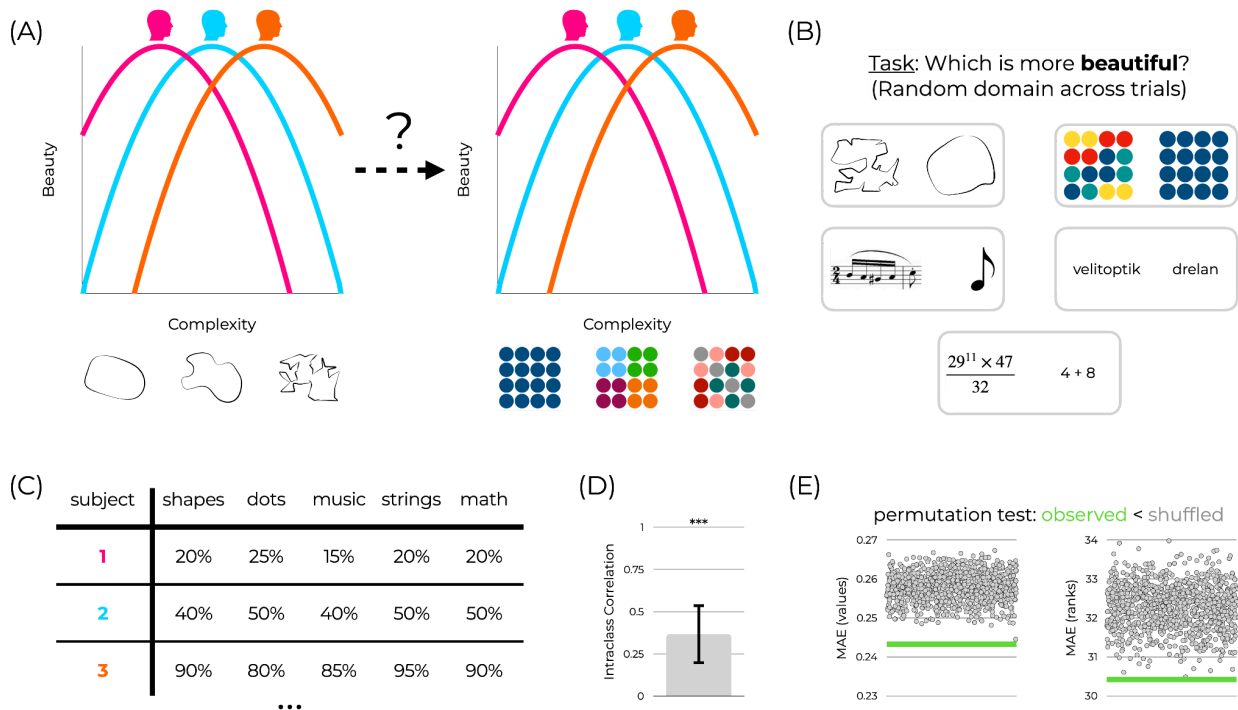


Fig 4. Individuals have stable complexity preferences across domains. (A) Previous work shows an inverse U-shape relationship between complexity and beauty. However, some people prefer more complex stimuli than other people (e.g., a different peak in their inverted U). Are such preferences stable within one person across different domains? (B) In Experiment 9, each trial presented a simple and complex stimulus from a single domain; subjects had to say which was more (or less) beautiful. Across trials, the domains varied randomly. (C) The resulting data index subject-level aesthetic preferences across domains (depicted here with idealized data). (D) An intraclass correlation revealed significant individual consistency across domains. In other words, people who preferred complex (or simple) stimuli in one domain preferred complex (or simple) stimuli in other domains. (E) A bootstrap analysis further examined the individual stability of these preferences across domains. The observed error of our data was considerably lower than the permuted error (i.e., shuffling within-domain, across-subjects) when considering both raw values (where the observed error was lower than the permuted error in 1000/1000 iterations) and ranks (where the observed error was lower in 999/1000 iterations).

General Discussion

The experiments reported here point towards domain-general representation of complexity in cognition, such that the mind appreciates a shared property across qualitatively different stimuli. We observed successful transfer of complexity across a variety of domains in ways that support reward-learning (Experiments 1–7), intrude on basic judgments (Experiment 8), and even underlie stable individual preferences (Experiment 9). In contrast to how complexity is typically studied in psychology (with different approaches and parameterizations for different stimuli), these results constitute evidence that the mind represents the shared complexity that unifies stimuli from otherwise disparate domains.

The logic of our approach mirrors recent work in other cognitive domains, such as numerical cognition. A longstanding research tradition analyzes the mechanisms, origins, and development of number perception, posing similar questions about the domain-generality of the relevant representations: When we see some number of dots and hear some number of tones (or touch some number of bumps, etc.), to what extent does the mind represent the single quantity shared across these domains? An illuminating series of discoveries answered this question by discovering transfer across these domains (Arrighi et al., 2014; Izard et al., 2009; Starkey et al., 1983; Starkey et al., 1990), much like how we find evidence for a common representation of complexity across domains. Accordingly, the approaches taken here may prove useful for the study of many different kinds of abstract representations across domains or stimuli, such as the abstract representation of ‘boundedness’ across events and objects (Papafragou & Ji, 2023), ‘extent’ across space and time (Yousif & Scholl, 2019), ‘symmetry’ across language and vision (Hafri et al., 2023), or even ‘randomness’ across different stimulus-formats (Boger et al., 2025).

Complexity and the language of thought

The present results connect to a recent body of experimental work concerning mental compression algorithms (Sablé-Meyer et al., 2021, Sablé-Meyer et al., 2022, for review see Dehaene et al., 2022), which has shed new light on longstanding debates about the format of mental representation. This work has proposed that the cognitive capacities which facilitate representation of geometric shapes, sequences of musical tones, and related stimuli rely on compressing the stimulus into a structured composition of its primitives. Such representations are proposed to take on a language-like format, deriving from a more general ‘language of thought’ (Fodor, 1975; Quilty-Dunn et al., 2023). The language of thought hypothesis posits that the format and structure of mental representations shares many properties with natural languages, such as containing discrete constituents that combine systematically to form structured representations, enabling flexible and generative thought.

While the same overarching language of thought may account for cognitive representations in any given domain (e.g., geometric shapes, auditory sequences), the primitives composing the language in each domain differ. The representation of a complex geometric shape consists of symbols like line segments and curves, while the representation of a complex sequence of

tones consists of symbols like pitches and amplitudes. By demonstrating successful transfer across a variety of domains, our work makes this connection more explicit. In other words, these representations not only take on a language-like structure but also interface seamlessly across domains.

Importantly, evidence *for* domain-general representation of complexity does not constitute evidence *against* the existence of any domain-specific representations of complexity. Indeed, many notions of complexity are definitionally domain-specific, as they cannot apply to other kinds of stimuli (e.g., curvature is a useful proxy for the complexity of a shape, but is not even defined when considering the complexity of a melody). Our proposal is only that, regardless of the nature and extent of any domain-specific representations of complexity in the mind, there is a domain-general notion of complexity in the mind that unifies stimuli across domains.

Origins and development

A natural question concerns the developmental origins of domain-general representations of complexity (also mirrored in the case of number; Feigenson et al., 2004). Previous work links complexity with exploration, demonstrating that children attend to and act on stimuli that are neither too simple nor too complex (Berlyne, 1950; Berlyne, 1966; Kidd et al., 2012; Kidd et al., 2014). Additionally, children may have an intuitive notion of *entropy*, noticing that non-agentive forces (e.g., the wind) can only increase entropy and disorder, whereas agentive forces (e.g., people) can either increase or decrease entropy and disorder (Newman et al., 2010). These and other results demonstrate that children represent some notion of complexity, but it remains open how stimulus-general that notion truly is. Would it extend across radically different domains in the ways we find here? Future work may use the same kinds of straightforward reward-transfer tasks (as in Experiments 1–7) to explore this question.

Along these same lines, a further question remains as to the phylogenetic origins of complexity representations. Recent work suggests that the capacity to represent shape complexity via a compression-like algorithm may be a uniquely human quality: Whereas humans distinguish shapes based on rich differences in the shapes' primitives (in ways captured by a symbolic model of shape), baboons distinguish shapes based on the shapes' visual differences alone (in ways captured by a Convolutional Neural Network; Sablé-Meyer et al., 2021). Given this recent work, it may be that non-human primates and other animals fail to represent stimulus complexity in a maximally abstract fashion, and may thus fail at the types of reward-transfer tasks we use here. On the other hand, other evidence suggests that primates possess abstract representations of other quantities such as number and its corresponding operations (Nieder, 2013; for review, see Dehaene et al., 1998), though the nature of these representations are still debated (Biro & Matsuzawa, 2001; Matsuzawa, 2009). Furthermore, recent evidence suggests that monkeys' representations of geometry (those that would be recruited in the complexity tasks used here) rely on similar cognitive processes to their representations of number (Cantlon et al., 2025; see also Dehaene et al., 2025). Thus, it is possible that similar kinds of mechanisms to those that support abstract representations of number may support abstract representations of complexity. Asking whether abstract representations of complexity exist in

non-human primate minds thus seems especially fruitful, regardless of the success or failure of the animals on these tasks.

Materials and Methods

The following sections contain in-depth descriptions of the experiments performed. The design, sample size, exclusion criteria, and analysis methods for every experiment reported here was pre-registered. These pre-registrations — along with stimuli, experiment scripts, data, and analysis scripts — are available at https://osf.io/42umv/overview?view_only=5dfac5c7d69641c7a63a7a8702d654b3. Readers can also experience every experiment for themselves (except for Experiment 7, which was conducted in-person) at https://perceptionresearch.org/complexity_transfer.

Subjects for Experiments 1–6, 8, and 9 were recruited online via Prolific (for a discussion of the reliability of this subject pool, see Peer et al., 2017). Subjects for Experiment 7 were recruited in-person at Johns Hopkins University. The experiments were approved by the Homewood Institutional Review Board of Johns Hopkins University.

Note that the reward-learning tasks (Experiments 1–7) are all nearly identical in design, procedure, and analysis; often only the stimuli varied across these experiments. Thus, for those experiments, all elements should be assumed to be the same as in Experiment 1 unless otherwise specified.

Experiment 1: Transfer between geometric shapes and dot-arrays

Experiment 1 tested bidirectional transfer between geometric shapes and dot-arrays in a reward-learning task. We asked whether subjects who learned that more (or less) complex shapes yielded greater rewards would spontaneously choose more (or less) complex dot-arrays (or vice versa) in a single, one-shot transfer trial.

Stimuli and procedure.

At the start of the experiment, subjects were randomly assigned to be rewarded for selecting either the simple stimulus or the complex stimulus on each learning trial. Additionally, the learning stimulus class was randomly assigned for each subject (either geometric shapes or dot-arrays, with the other stimulus class being the transfer stimulus).

Subjects completed 12 learning trials. On each trial, subjects chose between two options on the screen (left or right) and received feedback regarding their choice. If subjects chose the correct option, they heard a ‘ding’ sound, and saw a virtual coin drop into a piggy bank, thus increasing the ‘coin bank’ by 5 points; if subjects chose the incorrect option, they heard a negative buzzer sound, and saw no change in the coin bank. After 12 learning trials, subjects completed a single transfer trial in which a new stimulus class was presented. This key transfer trial was the main target of our analyses.

The geometric shapes used in this experiment were drawn from Sun & Firestone (2022a). The dot-array stimuli were inspired by Sun & Firestone (2022b), but were generated anew for this experiment.

Analyses and results.

As per our pre-registered analysis plans, subjects were excluded if they did not submit a complete dataset, or if they earned the reward on less than 50% (6 of 12) learning trials. Additionally, we excluded trials with a response time below 200ms. 101 subjects completed the task (one subject more than our pre-registered sample size participated, due to a feature of Prolific's recruiting process), 6 of which were excluded by our criterion for the learning trials. Of the remaining 1235 trials, 5 were excluded by the response time criterion.

86.3% of subjects transferred the reward rule to the new, unseen stimulus class ($p < .001$ in exact binomial test).

Experiment 2: Transfer between geometric shapes and melodies

Experiment 2 used the same design as Experiment 1, but used melodies instead of dot-arrays. Thus, this experiment tested bidirectional transfer across the visual and auditory modalities.

Stimulus and procedure.

The experiment's design was identical to that of Experiment 1. The only difference was that the melodies replaced the dot-arrays. On those trials, the two options were presented as music notes on each side of the screen. Clicking on a music note would play a melody; after listening to both melodies, subjects chose which one they thought would earn a reward. The simple melodies contained only a single note, while the complex melodies contained multiple notes with varying tempos and pitches. Melodies were generated in Ableton.

Analyses and results.

1 subject was excluded for not submitting a complete dataset, and 3 additional subjects were excluded for failing to earn the reward on at least 50% of learning trials. 1 of the remaining 1248 trials was excluded for speed.

71.9% of subjects transferred between the geometric shapes and the melodies ($p < .001$ in exact binomial test).

Experiment 3: Transfer between geometric shapes and letter-strings

Experiment 3 tested bidirectional transfer between geometric shapes and letter-strings.

Stimulus and procedure.

For the letter-strings, we drew from (and added to) the stimuli in Lewis & Frank (2016). Complex letter-strings were 4 syllables long, while simple letter-strings were 2 syllables long. To reach 12 simple and 12 complex letter-strings, we generated additional strings that fit these restrictions.

Analyses and results.

101 subjects completed the task (one subject more than our pre-registered sample size participated, due to a feature of Prolific's recruiting process). 10 were excluded by our learning phase criterion; 22 of the remaining 1183 trials were excluded for speed.

78.0% of subjects transferred between the geometric shapes and the letter-strings ($p < .001$ in exact binomial test).

Experiment 4: Transfer between geometric shapes and mathematical expressions

Experiment 4 tested bidirectional transfer between geometric shapes and mathematical expressions.

Stimulus and procedure.

The simple and complex mathematical expressions within each trial contained identical terms, and only differed in their operations. Both expressions contained 4 unique integers between 11 and 49. The integers were randomly generated, while ensuring they were not divisible by 10 and were not multiples of each other. The simple mathematical expression merely added the four terms together, while the complex one included a variety of operations (namely, exponentiation, multiplication, and division). So, for example, a simple expression might be $17 + 13 + 24 + 31$, while the accompanying complex expression would be $\frac{17^{13} \times 24}{31}$.

Analyses and results.

100 subjects completed the task. 2 subjects were excluded for failing to learn the reward-rule, and 3 of 1274 remaining trials were excluded for speed.

64.3% of subjects transferred the reward-rule between the geometric shapes and the mathematical expressions ($p < .01$ in exact binomial test).

Experiment 5: Transfer between all pairs of previous stimuli

Experiment 5 tested whether transfer would arise between *all* of the stimuli used in previous experiments. In other words, the previous experiments tested bidirectional transfer between geometric shapes and a host of other stimuli; but this leaves open the question of whether transfer arises between any pair of stimuli (and not just ones that include geometric shapes).

Stimuli and procedure.

This experiment contained 5 possible stimulus classes, each of which were used in Experiments 1–4 (geometric shapes, dot-arrays, melodies, letter-strings, and mathematical expressions). A random stimulus class was selected for the learning phase, and then the transfer phase consisted of a single trial (without feedback) of each of the 4 held-out stimulus classes. As before, the learning stimulus was chosen randomly for each subject, as was the reward rule (i.e., reward-simple or reward-complex). In this way, the design of this experiment is identical to Experiments 1–4, except that the transfer phase now consisted of 4 trials (1 for each held-out stimulus class) instead of 1.

Analyses and results.

Of the 499 subjects who submitted full data for this experiment, 40 were excluded by the learning phase criterion. 13 of the remaining 7344 trials were excluded for speed.

Our first pre-registered analysis tested whether successful transfer arose *from* each stimulus class. Thus, we conducted *t*-tests to ask whether the mean transfer accuracy across all 4 transfer trials was above-chance for subjects who were presented with any of our stimulus classes in the learning phase. In each case, we observed above-chance transfer. The mean transfer accuracy for each class was as follows: shapes: 74.3%, $t(111) = 9.56$, $p < .001$, $d = 0.90$; dot-arrays: 68.6%, $t(81) = 5.84$, $p < .001$, $d = 0.65$; melodies: 81.0%, $t(80) = 11.35$, $p < .001$, $d = 1.26$; letter-strings: 65.8%, $t(90) = 4.83$, $p < .001$, $d = 0.51$; mathematical expressions: 61.0%, $t(92) = 3.62$, $p < .001$, $d = 0.38$.

Our second pre-registered analysis asked whether successful transfer arose *to* each stimulus class. Thus, we conducted exact binomial tests comparing the mean transfer accuracy for any given transfer stimulus to chance. We again observed above-chance transfer in every case: shapes: 78.0%, $p < .001$; dot-arrays: 73.7%, $p < .001$; melodies: 63.8%, $p < .001$; letter-strings: 65.5%, $p < .001$; mathematical expressions: 70.4%, $p < .001$.

Our final pre-registered analysis asked whether successful transfer arose *between* each stimulus pair. Thus, we conducted 10 binomial tests on transfer accuracy, one for each possible learning-transfer pair. We observed significantly above-chance transfer in 9/10 of these pairs, and marginally significant transfer in the final pair: dot-arrays and mathematical expressions: 64.0%, $p < .001$ in exact binomial test; dot-arrays and melodies: 75.5%, $p < .001$; dot-arrays and shapes: 82.5%, $p < .001$; dot-arrays and letter-strings: 62.2%, $p < .01$; mathematical expressions and melodies: 65.3%, $p < .001$; mathematical expressions and shapes: 75.6%, $p < .001$; mathematical expressions and letter-strings: 56.5%, $p = .09$; melodies and shapes: 74.1%, $p < .001$; melodies and letter-strings: 72.1%, $p < .001$; shapes and letter-strings: 71.8%; $p < .001$.

Experiment 6: Transfer for complexity, but not color

Experiment 6 asked whether transfer would arise for merely any stimulus dimension, and not just those having to do with complexity. To test this, we re-used the design of Experiment 1 but added a key color transfer trial. Notably, colors can vary in a stimulus dimension (here, saturation) that is not necessarily related to complexity. Thus, we predicted that no transfer would arise to color dimensions.

Stimulus and procedure.

This design exactly replicated that of Experiment 1, but added one key transfer trial. In this additional transfer trial, subjects were presented with two colors of identical hue and lightness, but of differing saturation. (The hue was chosen randomly for each subject, and the lightness was always 50%). The more saturated color had 75% saturation, compared to the less saturated color's 25%. Importantly, colors could never be chosen as the learning stimulus; the learning stimulus could only be geometric shapes or dot-arrays, as in Experiment 1.

Analyses and results.

99 subjects submitted full data, and 3 subjects were excluded due to the learning phase criterion. 20 of the remaining 1344 trials were excluded for speed; one of these trials was a transfer trial, which explains why the degrees of freedom for the paired t -test that follows is 94 (as opposed to 95).

First, we observed that subjects transferred between the geometric shapes and dot-arrays, as they did in Experiment 1 (at a rate of 84.4%, $p < .001$ in exact binomial test). However, subjects did *not* transfer from either of these stimuli to the color trial; in other words, performance was indistinguishable from chance on the color transfer trials (with 48.4% of subjects earning the reward, $p = 0.84$ in exact binomial test). Additionally, subjects were significantly more accurate on the non-color (i.e., geometric shape or dot-array) transfer trial than they were on the color transfer trial (mean difference = 35.8%; $t(94) = 5.99$, $p < .001$, $d = 0.62$).

Experiment 7: Transfer between tactile forms and dot-arrays

Experiment 7 tested transfer to a new modality: touch. Here, we asked whether bidirectional transfer would arise between tactile forms (which participants had no visual access to) and dot-arrays as in previous experiments.

Stimuli and procedure.

We used the same design here as in the other transfer experiments: Participants completed 12 learning trials, followed by 1 transfer trial. The dot-arrays were the same ones used in previous experiments. We 3D printed 12 simple shapes and 12 complex shapes (selected from the shapes used in previous experiments) with a Bambu Lab X1C, and used those as our tactile forms.

Participants were recruited in-person at Johns Hopkins University. For the tactile trials, the experimenter placed one shape in each of two cardboard boxes. Each box contained holes in both sides, allowing participants to touch the shapes. However, participants had no visual access to the shapes; thus, they were making their judgments purely based on tactile information. As in the online experiments, participants received auditory feedback indicating whether or not they earned a reward on each turn. For the dot-array trials, participants used the experimenter's computer to input their responses.

Analyses and results.

We recorded data from 99 subjects (a server error prevented data recording for 1 additional subject). 0 subjects were excluded for responding correctly on less than 50% of the learning phase. 4 trials were excluded for speed.

81.8% of subjects transferred the reward-rule between the tactile forms and the dot-arrays ($p < .001$ in exact binomial test).

Experiment 8: Automatic interference in a Stroop-like task

Experiment 8 tested whether representations of complexity in one domain automatically intrude on basic judgments of complexity in another domain.

Stimuli and procedure.

We designed a modified Stroop task. On each trial, subjects were presented with two words — one simple and one complex — and simply had to choose which one was more complex in one block, and more simple in another. (The order of the two blocks was randomized for each subject; we used a blocked design to ensure that any observed effects did not merely arise because of a preference for choosing the complex or simple words.) Each block contained 60 trials. The words used were the same as those used in Experiment 4, with the addition of several new words that followed the same rule as in that experiment (i.e., complex words have 4 syllables, simple words have 2 syllables).

Importantly, above each word, a task-irrelevant shape appeared (from the shapes used in Experiment 1). Participants were instructed only to judge the words, and that they could totally ignore the shapes. On one half of trials, a complex shape appeared above the complex word, and a simple shape appeared above the simple word — i.e., shape complexity was congruent to word complexity. On the other half of trials, a complex shape appeared above the simple word, and the simple shape appeared above the complex word — i.e., shape complexity was incongruent to word complexity. Each block contained an equal number of congruent and incongruent trials, and the order of these trials was randomly shuffled for each participant.

Analyses and results.

1 subject did not submit a full dataset, and 1 additional subject was excluded by our pre-registered accuracy criterion (which excluded subjects who gave the correct answer on less than 75% of trials). Of the 11520 trials remaining, 456 (3.96%) were excluded by our pre-registered response time criterion (which excluded trials with a response time below 200ms or above 1500ms).

As one would expect if representations of complexity interfere with basic judgments across domains, subjects were faster on congruent shape-word trials than incongruent shape-word trials (57.4ms difference, $t(47) = 9.07$, $p < .001$, $d = 1.31$ in paired t -test). And this was not just the result of a speed-accuracy tradeoff; in fact, subjects were also more accurate on congruent trials (mean difference: +2.49%, $t(47) = 3.26$, $p < .01$, $d = 0.47$).

Experiment 9: Stable individual differences in aesthetic judgments

Experiment 9 tested whether domain-general representations of complexity govern stable individual differences in aesthetic judgments across those domains.

Stimuli and procedure.

On each trial of this experiment, participants were presented with a simple and complex stimulus of a given type. In one block, participants were asked to choose which option they found more beautiful; in the other block, they were asked to choose which option they found less beautiful. (The purpose of this blocked design was to ensure that any observed effects do not arise due purely to a preference to choose the complex option. The order of the two blocks was randomized for each subject.) Each trial presented only one stimulus class, but across trials the stimulus classes varied randomly; i.e., one trial might depict two geometric shapes, while the next might contain two mathematical expressions. We used the 5 stimulus classes from Experiment 5 (geometric shapes, dot-arrays, melodies, letter-strings, and mathematical expressions). Each block contained 30 trials, allowing for each of the 5 stimulus classes to appear 6 times. Trial order within a given block was randomized for each participant.

Analyses and results.

All 150 subjects submitted a full dataset, and we pre-registered no subject-level exclusion criteria. 48 of the 9000 trials were excluded based on our pre-registered response time exclusion criterion (which excluded trials with a response time below 200ms).

First, we computed an intraclass correlation coefficient, which revealed consistent individual differences in complexity-driven aesthetic preferences across domains (two-way ICC(C,5) = 0.37, $F(149, 596) = 1.58$, $p < .001$).

Next, we conducted two bootstrap analyses of this reliability. These tests allowed us to better understand the observed stability relative to chance. The basic intuition for these tests is to

compare the observed within-subject consistency to the level of consistency that would arise after shuffling all preferences within-domain, across-subjects. So, we conducted a simulation in which, in each of 1000 iterations, we shuffled the values within-domain, across-subjects. Then, on these shuffled values, we computed a new 'mean absolute error' — i.e., the difference between a subject's preference in a given domain and their average preference across domains. We compared these values to the observed mean absolute error in our data. First, we performed this analysis for the raw values (i.e., how often a subject found the complex option more beautiful within a domain); this revealed a high level of stability, as our observed error was lower than the permuted error in 1000/1000 iterations. Second, we performed this analysis for ranked values; here, our observed error was lower than the permuted error in 999/1000 simulations. Thus, these results point to the stability of complexity-driven aesthetic preferences across domains.

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